Areas Related to Circles

Recap

- Segment: area between chord and corresponding arc
- Sector: Area between Radii and corresponding arc

Area of sector of angle θ $\quad C(O, r)$

$\rightarrow \frac{\theta}{360} \times \pi r^2$

Diameter = PQ

Centre =

- Area of 2D shapes:

area of Square = ar(□) = side^2

ar (Rect) = $l \times b$

ar (Rhombus) = $d_1 \times d_2$

ar (||gm) = $b \times$ corresponding hieght

ar (Trapezium) = $1/2 \times h$ (at b)

Peri (Sq) = 4(side)

ar ⬡ = $\sqrt{3}/4 \times a^2$

- Properties -

① In a square and Rectangle diagonals are equal.

① In Rhombus and square diagonals bisect each other.

Introduction -

→ Complete angle = 360°

→ Area of sector :

If angle is - 360° = πr^2

$1 = \frac{\pi r^2}{360°}$

$\theta = \frac{\pi r^2}{360°} \times \theta$

→ Area of quadrant → $\angle = 90°$

$\frac{1}{4} \pi r^2$ or $\frac{90°}{360} \times \pi r^2$

- ar (8 slices) = $\frac{1}{8} \pi r^2 = \frac{1}{4} \pi r^2$

- length of the wire used for gold brooch

$= 2\pi r + 4d$

- length of an arc of a sector of $\angle \theta$

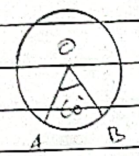
Total length = $2\pi r$ ($\angle = 360°$)

if $\angle = 1$ then length = $\frac{2\pi r}{360}$

If $\angle = \theta$ then length = $\frac{2\pi r}{360}$

Ex-11.1

1-Solu Radius of the circle = 6 cm (= r)
Central angle = 60° (= θ)



∴ Area of the Sector = $\frac{\pi r^2 \theta}{360}$

$= \frac{22}{7} \times \frac{6 \times 6 \times 60}{360} = \frac{132}{7}$ cm² = 18.86 cm²

2-Solu Circumference of the circle = $2\pi r$ = 22 cm

$2\pi r = 22$

$r = \frac{22}{2\pi} = \frac{22 \times 7}{2 \times 22} = \frac{7}{2}$ cm

Central angle of quadrant = 90° (= θ)

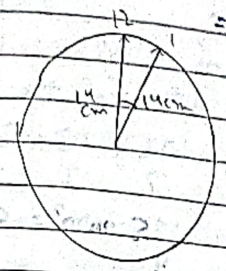
∴ Area of a quadrant = $\frac{\pi r^2 \theta}{360°}$

$= \frac{\frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 90}{360} = \frac{77}{8}$ cm² =

= 9.625 cm²

Hence, the area of the quadrant is 9.625 cm².

5-Solu Radius of the circle = length of minute hand = 14 cm



We know that minute hand covers the full circle in 60° minutes.

∴ Angle covered in 5 minutes

$= \frac{360}{60} \times 5 = 30°$

Thus, angle of sector = 30°

∴ Area swept by minute hand in 5 minutes

$= \frac{\pi r^2 \theta}{360} = \frac{22}{7} \times 14 \times 14 \times \frac{30°}{360°}$

$= \frac{1}{12} \times 22 \times 28 = \frac{154}{3} = 51.33$ cm²

Hence, the area swept by minute hand in 5 minutes is 51.33 cm².

9-Solu Diameter of the circle = 35 mm
Radius of the circle = $\frac{35}{2}$ mm

Number of Diameters = 5
Number of equal sectors = 10
(i) total length of the sector wire
= length of 5 diameters + circumference
$= 5(35) + 2\pi r = 175 + 2 \times \frac{22}{7} \times \frac{35}{2}$

= 175 + 110 = 285 mm

(ii) Sector angle of broach

$$= \frac{360^\circ}{\text{Number of sectors}} = \frac{360^\circ}{10} = 36^\circ$$

Area of each sector (of the broach)

$$= \frac{\pi R^2 \theta}{360} = \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} \times \frac{36}{360}$$

$$= \frac{11 \times 35}{4} = \frac{385}{4} \text{ mm}^2 = 96.25 \text{ mm}^2$$

10-Soln Radius of circle = 45 cm

Number of ribs = 8

Central angle $(\theta) = \frac{360^\circ}{8} = 45^\circ$

$\therefore$ Area of the sector $= \frac{\pi R^2 \theta}{360}$

$$= \frac{22}{7} \times 45 \times 45 \times \frac{45}{360}$$

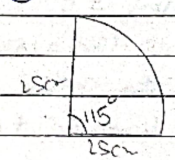
$$= \frac{22}{7} \times 45 \times 55 \times \frac{1}{8}$$

$$= \frac{12275}{28} \text{ cm}^2 = 795.53 \text{ cm}^2$$

Hence, the area between two consecutive ribs of the umbrella = 795.53 cm^2

11-Soln Length of blade = 28 cm

Sector angle = 115°



$\therefore$ Area of the sector = Area swept by one blade

$$= \frac{\pi R^2 \theta}{360} = \frac{22}{7} \times 25 \times 25 \times \frac{115}{360} = 627.48 \text{ cm}^2$$

Hence, the area covered by two blades

$$= 2 \times 627.48 = 1254.96 \text{ cm}^2$$

12-Soln Radius of sector = 16.5 Km

Central angle = 80°

$\therefore$ Area of the sector $= \frac{\pi R^2 \theta}{360}$

$$= \frac{3.14 \times 16.5 \times 16.5 \times 80}{360}$$

$$= 189.97 \text{ Km}^2$$

Hence, the area of the sea over which the ships are warned is 189.97 Km^2.

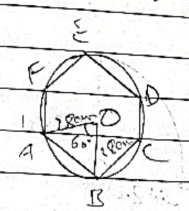
12-Soln Number of equal designs = 6

Radius of designs = 28 cm

Central angle of each design $= \frac{360}{6} = 60^\circ$

Clearly, ΔAOB is equilateral

[$\because OA = OB$ & $\angle AOB = 60^\circ$]



Area of the shaded design portion

= Area of the segment

= Area of the sector OAB − Area of ΔOAB

$$= \frac{\pi R^2 \theta}{360} - \frac{\sqrt{3}}{4} (\text{side})^2$$

$= \frac{22}{7} \times 28 \times 28 \times \frac{60}{360} - \frac{\sqrt{3}}{4} \times 28 \times 28$

$= 410.66 - 333.2 = 77.46 \text{ cm}^2$

$\therefore$ Area of the shaded designed portion.

$= 6(77.46) = 464.76 \text{ cm}^2$

Cost of making $1 \text{ cm}^2 =$ ₹ 0.35

Hence, the cost of making 464.76 cm^2

$= 0.35 \times 464.76 =$ ₹ 162.67.

143-Ans (d) $\frac{\rho}{720} \times 2\pi R^2$

Justification: Radius of circle = R

Sector angle $= \rho^\circ$

Area of sector $= \frac{\pi R^2 \theta}{360} = \frac{\pi R^2 \rho}{360}$

$= \frac{\rho}{720} \times 2\pi R^2$.